

# SCION: The Structural Intelligence Engine

Transitioning Teradata DNA from Rented Capability to Owned Intelligence.



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Managed Services

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# AGENDA



**SCION Overview**



**Scope & Architecture**



**Estimation** *(Resources, Infrastructure, Maintenance)*



**Consolidated Summary**



**Final Recommendations**

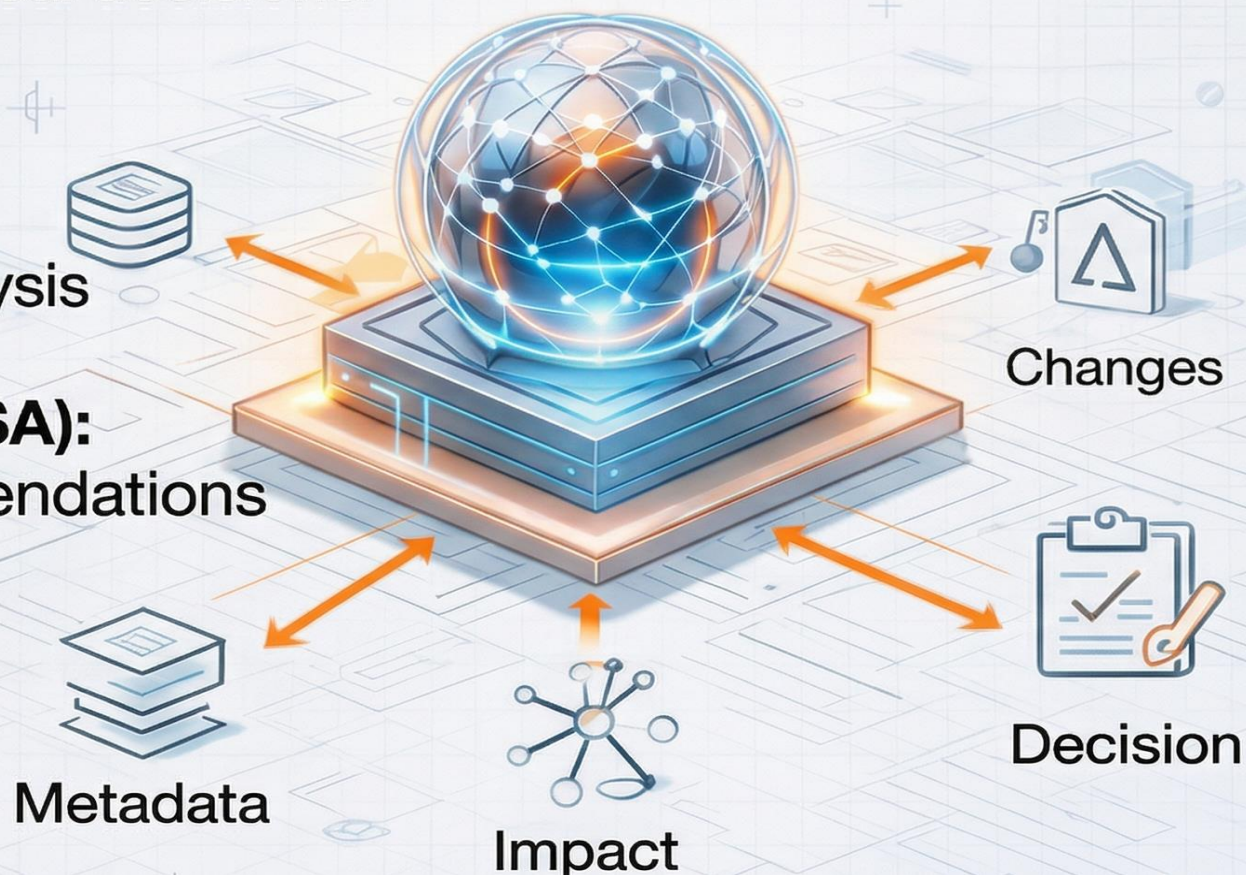
# SCION Overview



# What is SCION?

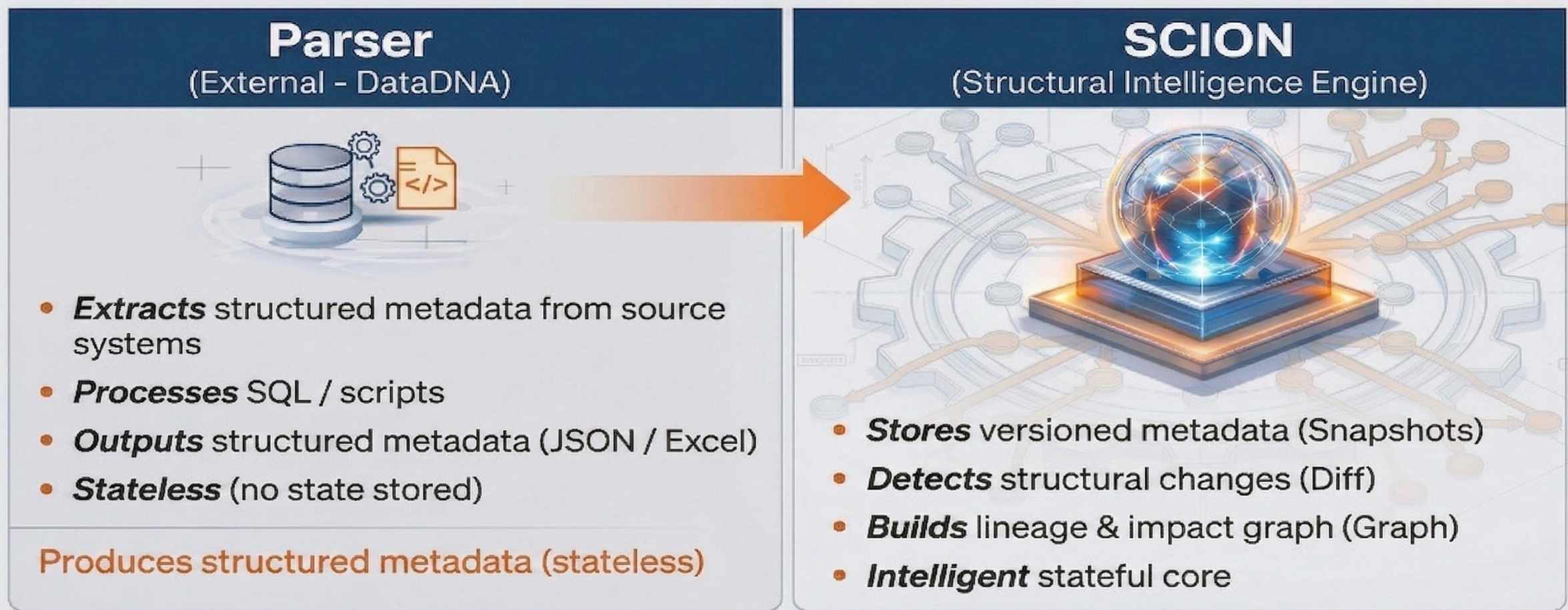
SCION is a Structural Intelligence Engine that transforms raw metadata into explainable, actionable technical decisions.

- **Snapshot + Diff + Graph** for metadata versioning, change detection, & impact analysis
- **Explainable AI reasoning (TAISA):** for risk evaluation and recommendations
- **Full traceability and control** over technical evolution.





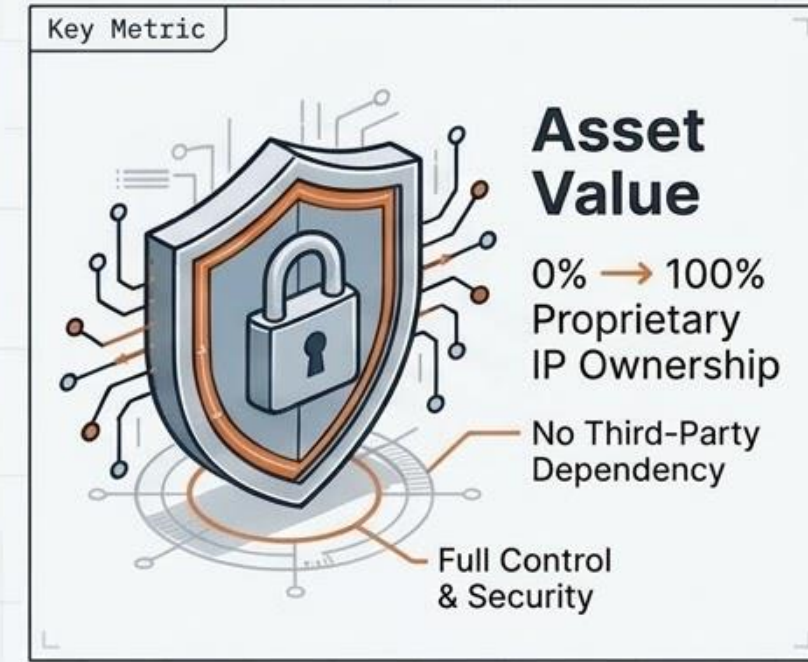
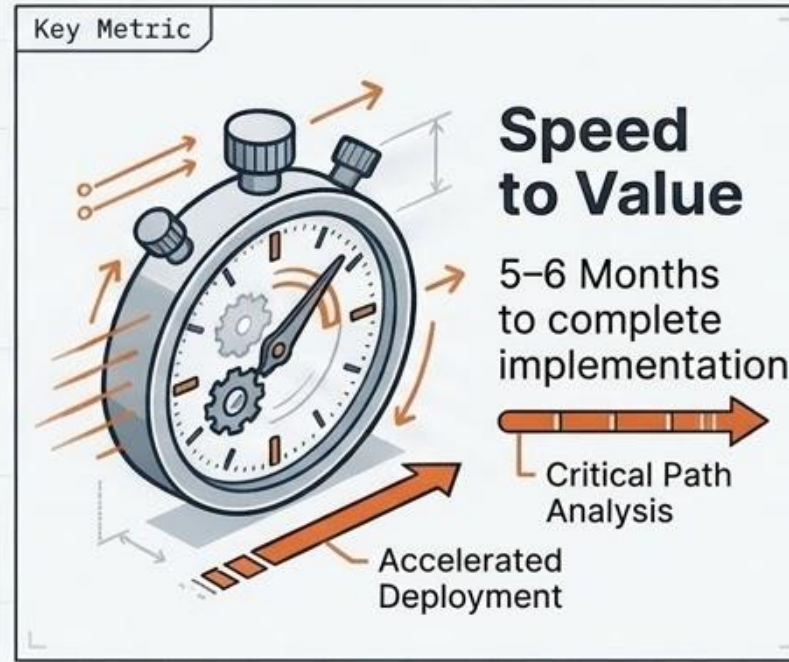
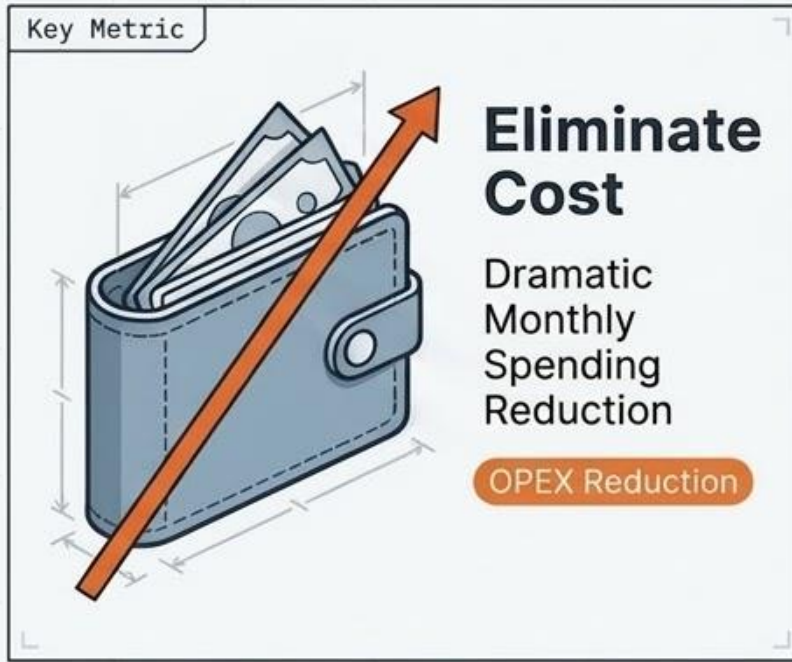
# End-to-End Flow: Parser → SCION → Insight



Aligned with **DataDNA** architecture: **Parser** performs **extraction** → **SCION** stores and analyzes



# The Case for Proprietary Governance



### EXECUTIVE TAKEAWAYS



**The Mandate:** Replace opaque ‘black-box’ legacy tools to eliminate exposure and recurring fees.



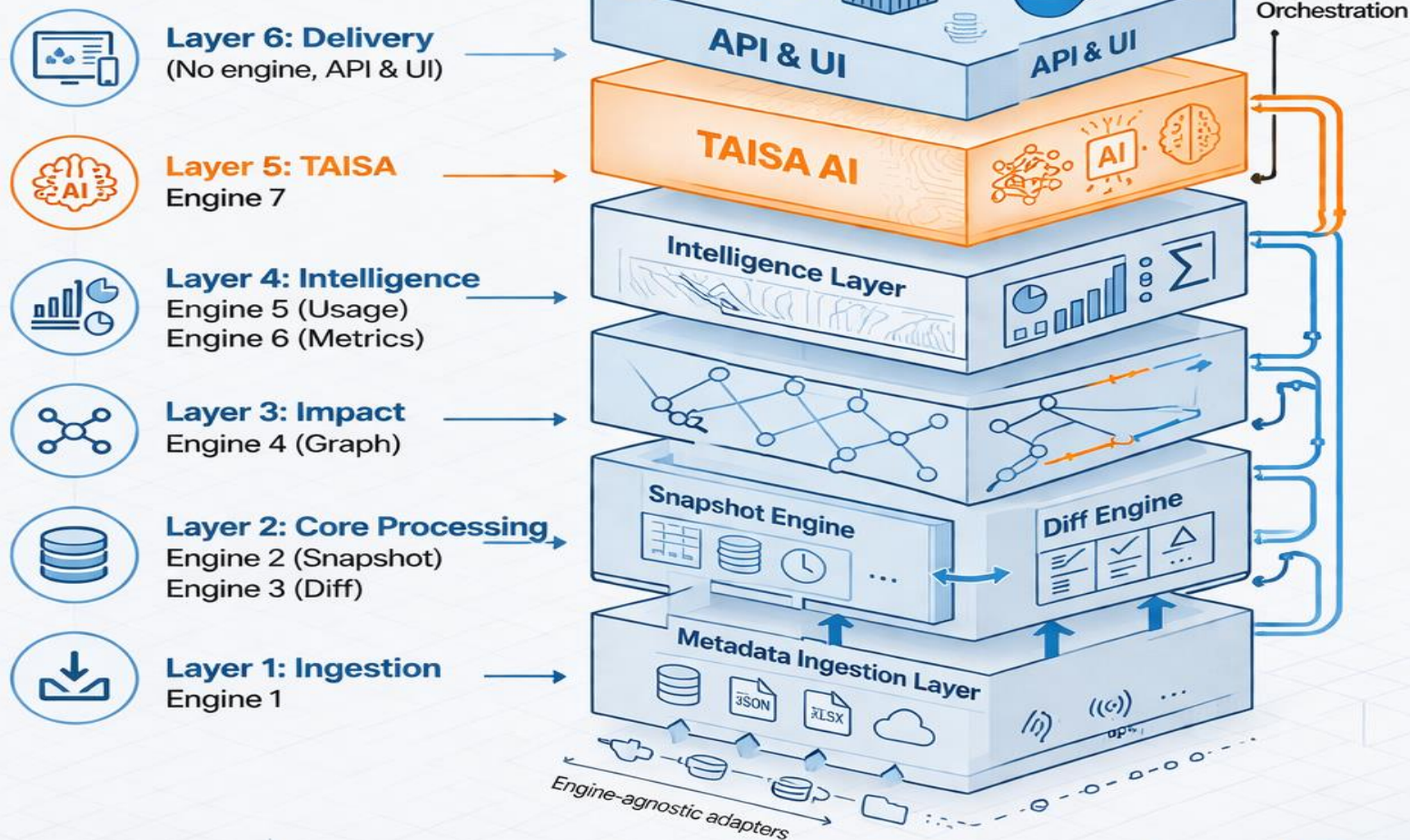
**The Solution:** Deploy SCION—a deterministic, modular governance platform with native AI augmentation.

# Scope and Architecture



# Architecture Overview: A Layered, Modular Design

- **Ingestion:** Parser-driven, stateless lineage derivation (JSON / Excel inputs).
- **Core Processing:** Deterministic state versioning and change detection via snapshots.
- **Impact Engine:** Applies parser-derived lineage using recursive graph logic.
- **Intelligence:** Usage-based weighting and volatility metrics.
- **TAISA AI:** Stateless insight generation from structured system events.
- **Delivery:** Dockerized runtime with secure API orchestration.





# Layer 1 - Engine 1: Metadata Ingestion Layer

## Role / Purpose:

The Universal Adapter. Responsible for extracting and normalizing structural metadata from EDW engines.

## Core Capabilities:

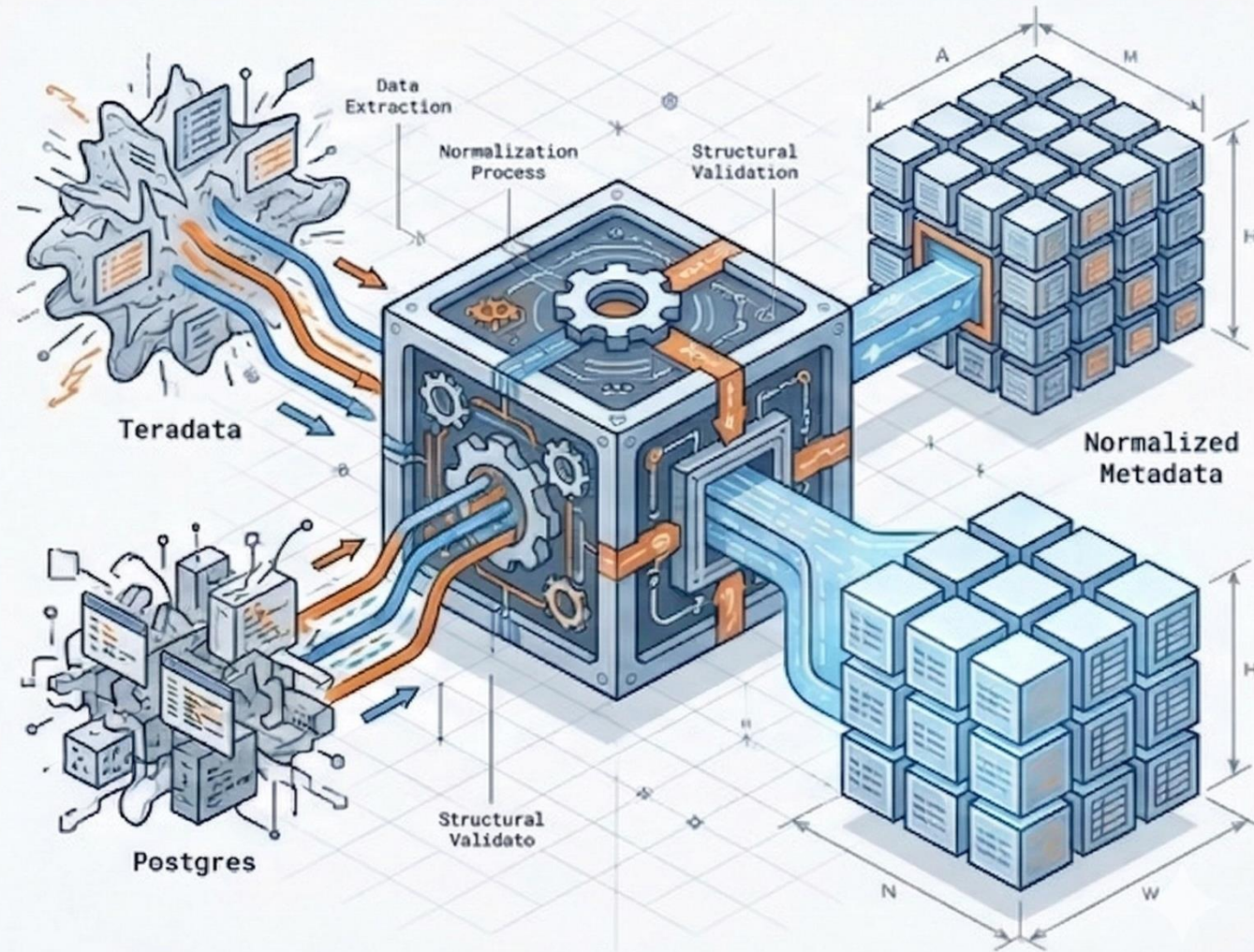
- Source-agnostic extraction (Teradata/Postgres adapters).
- Canonical **YAML-based template engine**.
- **Structural validation** and **typological normalization**.

## Technical Value:

Decouples governance from specific database vendors; eliminates raw catalog replication.

## Strategic Value:

Future-proofs the platform, allowing DNA to adapt to new EDW technologies without re-platforming.





# Layer 2 - Engine 2: Snapshot Engine

## Role / Purpose:

The Time Machine. Captures full EDW state to guarantee transactional consistency.

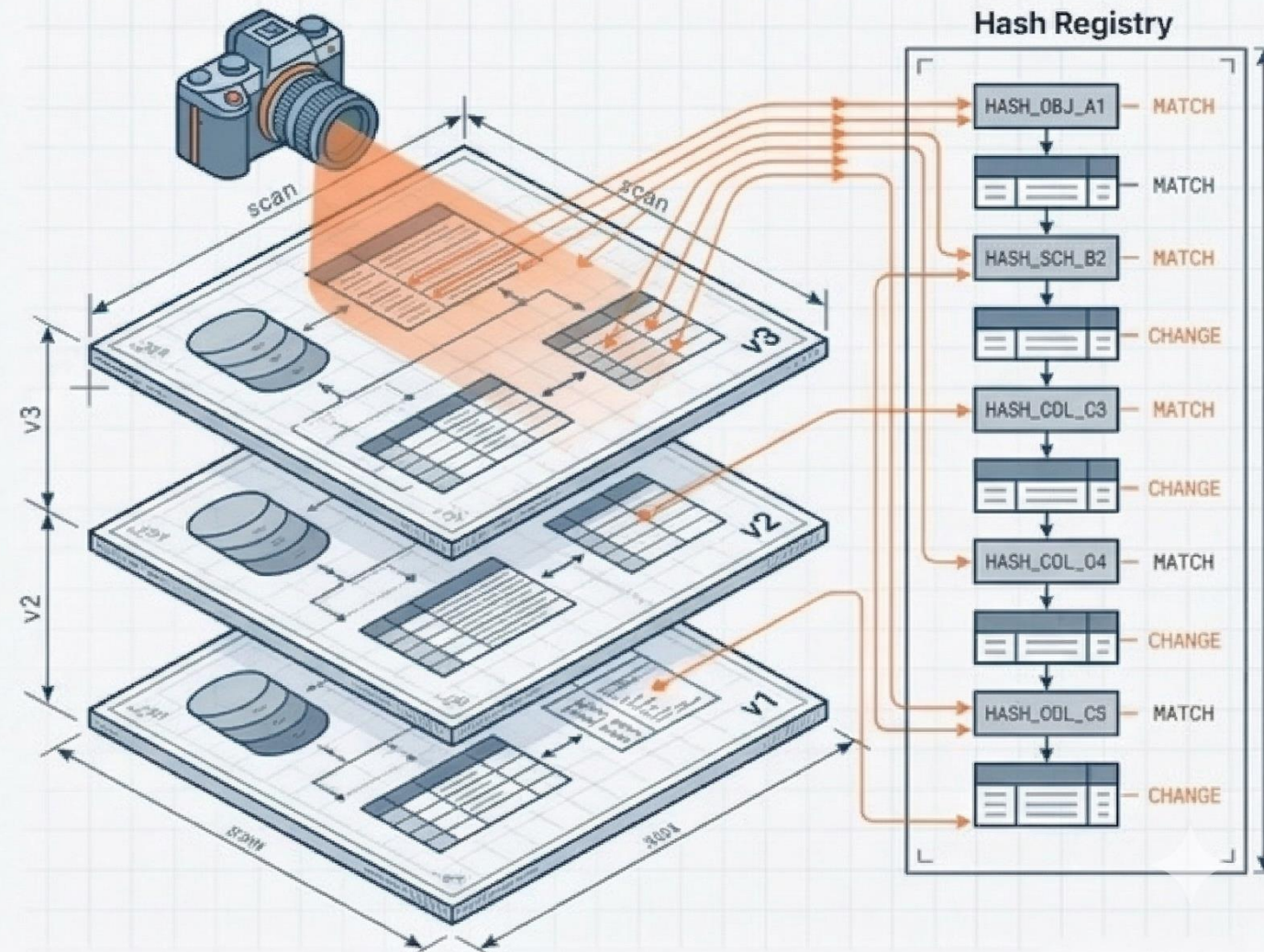
## Core Capabilities:

- Historical versioning with structural hashing.
- Detects changes via hash comparison (Zero data read).
- Metrics capture: Object, Schema, and Column counts.

**Technical Value:** Moves from passive tracking to active state monitoring.

## Strategic Value:

Provides the foundational 'truth' required for regulatory compliance and auditability.





# Layer 2 - Engine 3: Diff Engine

## Role / Purpose:

The Change Detective. Compares snapshots to identify and classify structural evolution.

## Core Capabilities:

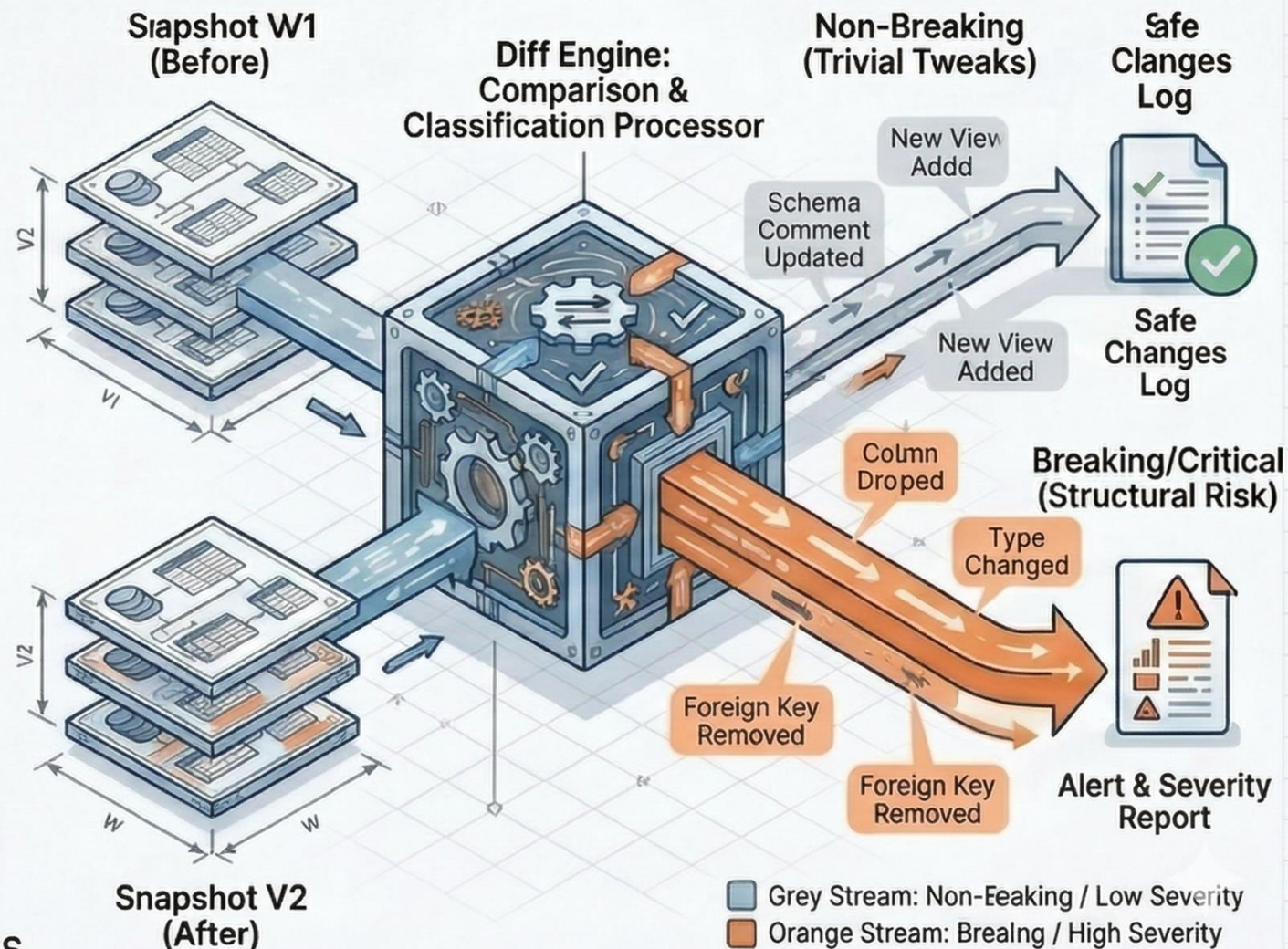
- Deterministic 'Before/After' comparison.
- 'Breaking' vs. 'Non-Breaking' classification.
- Severity scoring and Change Grouping by domain.

## Technical Value:

Consistent processing; produces audit-ready JSON logs of exactly *what* changed. (Using JetBrains Mono for 'JSON')

## Strategic Value:

Eliminates 'alert fatigue' by distinguishing between trivial tweaks and critical structural risks.





# Layer 3 - Engine 4: Graph & Impact Engine

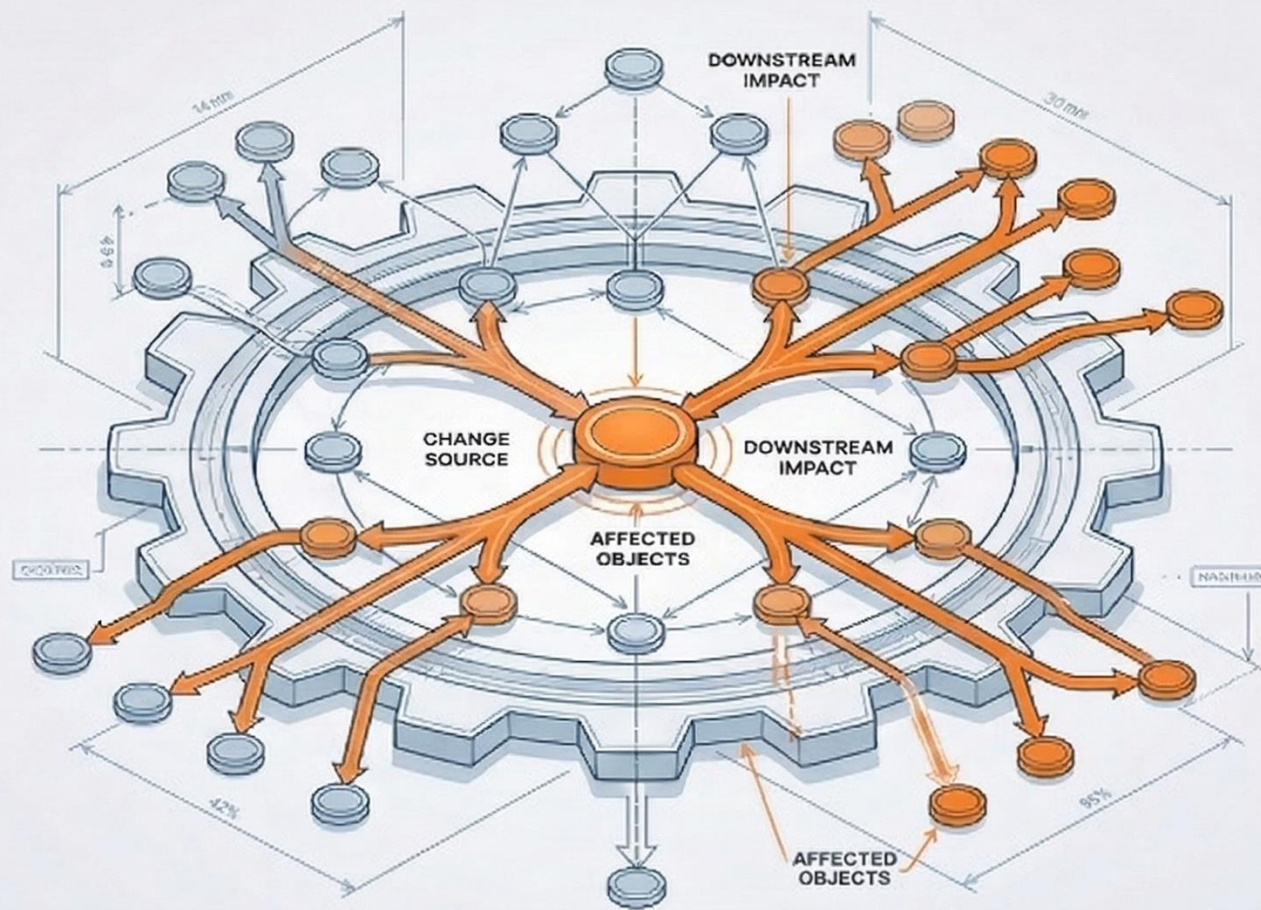
**Role / Purpose:** The Network Mapper. Consumes *parser-derived lineage* and calculates the “blast radius” of changes.

**Core Capabilities:**

- Builds versioned dependency graph from *parser-derived lineage* (Nodes & Edges).
- Scores fragility and Centrality metrics.
- Cycle detection and Critical Path analysis.

**Technical Value:** Native SQL graph logic removes dependency on external graph databases.

**Strategic Value:** Proactively predicts outages by visualizing downstream impact before deployment.





# Layer 4 - Engine 5: Usage & Criticality Engine

Engineering Transparency

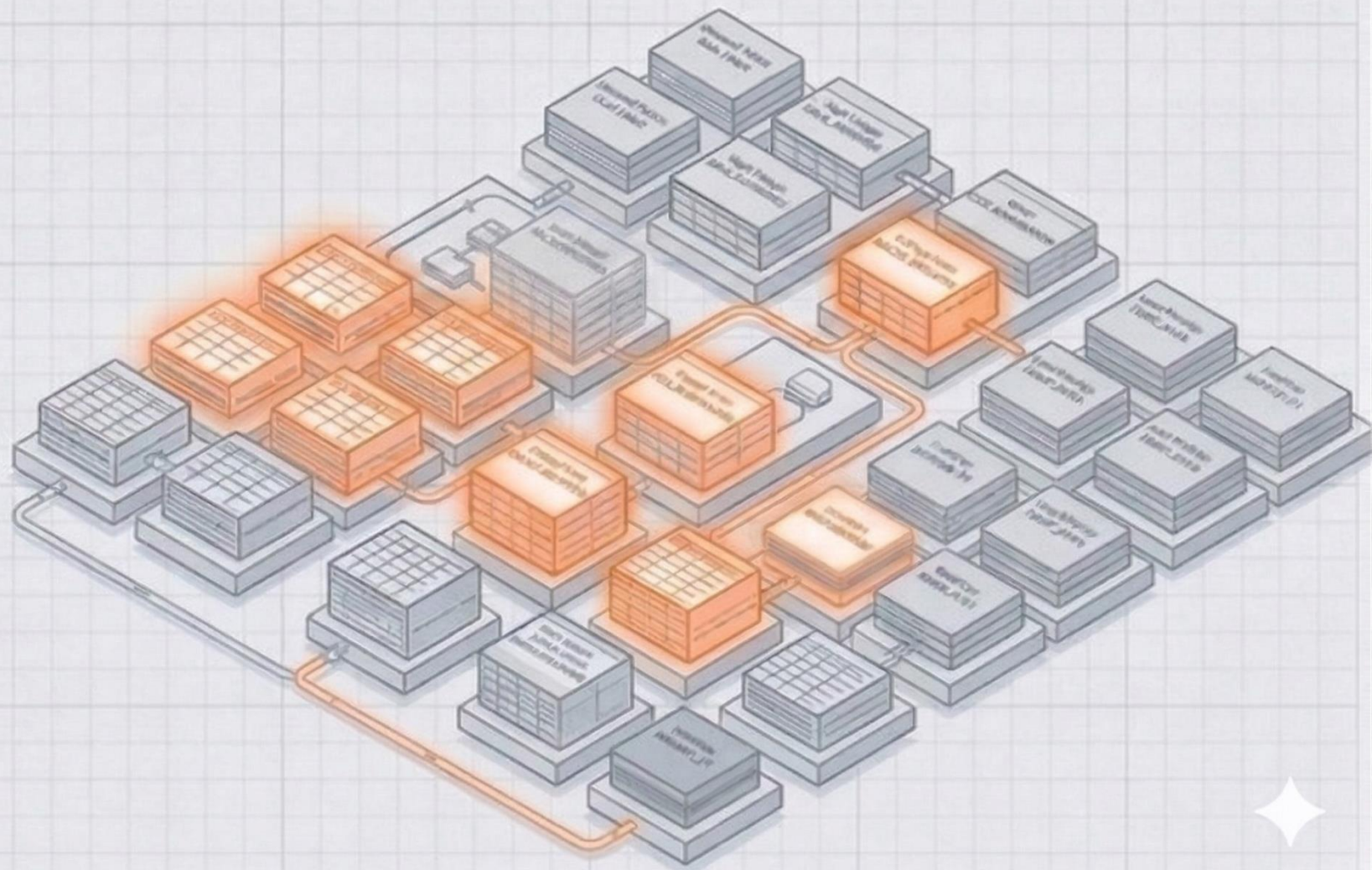
**Role / Purpose:** The Contextualizer.  
Weights technical structure with actual business usage reality.

**Core Capabilities:**

- Input from log parser.
- Usage frequency heat-mapping.
- User-weighting (e.g., CEO Report > Ad-hoc query).

**Technical Value:** Merges static metadata (structure) with dynamic logs (runtime).

**Strategic Value:** Shifts governance from theoretical impact to actual business risk.





# Layer 5 - Engine 6: Intelligence Metrics Layer

## Role / Purpose:

The Analyst. Aggregates raw signals into high-level ecosystem health indicators.

## Core Capabilities:

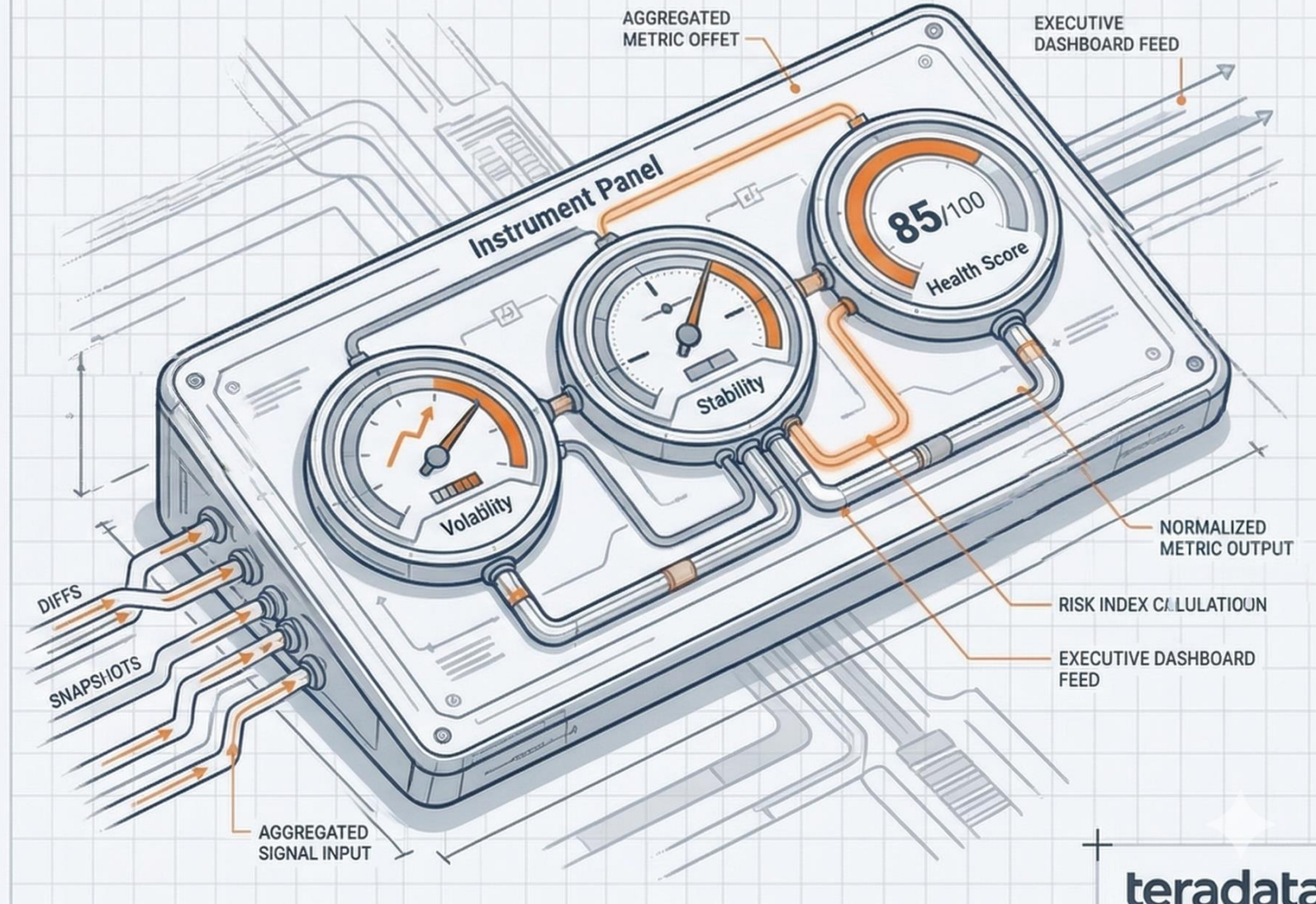
- **Volatility Indexing** (identifying unstable domains). (JetBrains Mono)
- **Stability Trends** per object. (JetBrains Mono)
- **Governance Scorecards** and **Domain Risk Indices**. (JetBrains Mono)

## Technical Value:

Transforms disparate data points (diffs, snapshots) into standardized metrics.

## Strategic Value:

Elevates SCION from a 'Metadata Tracker' to an 'Enterprise Structural Intelligence Platform'.





# Layer 6 - Engine 7: TAISA Integration Layer

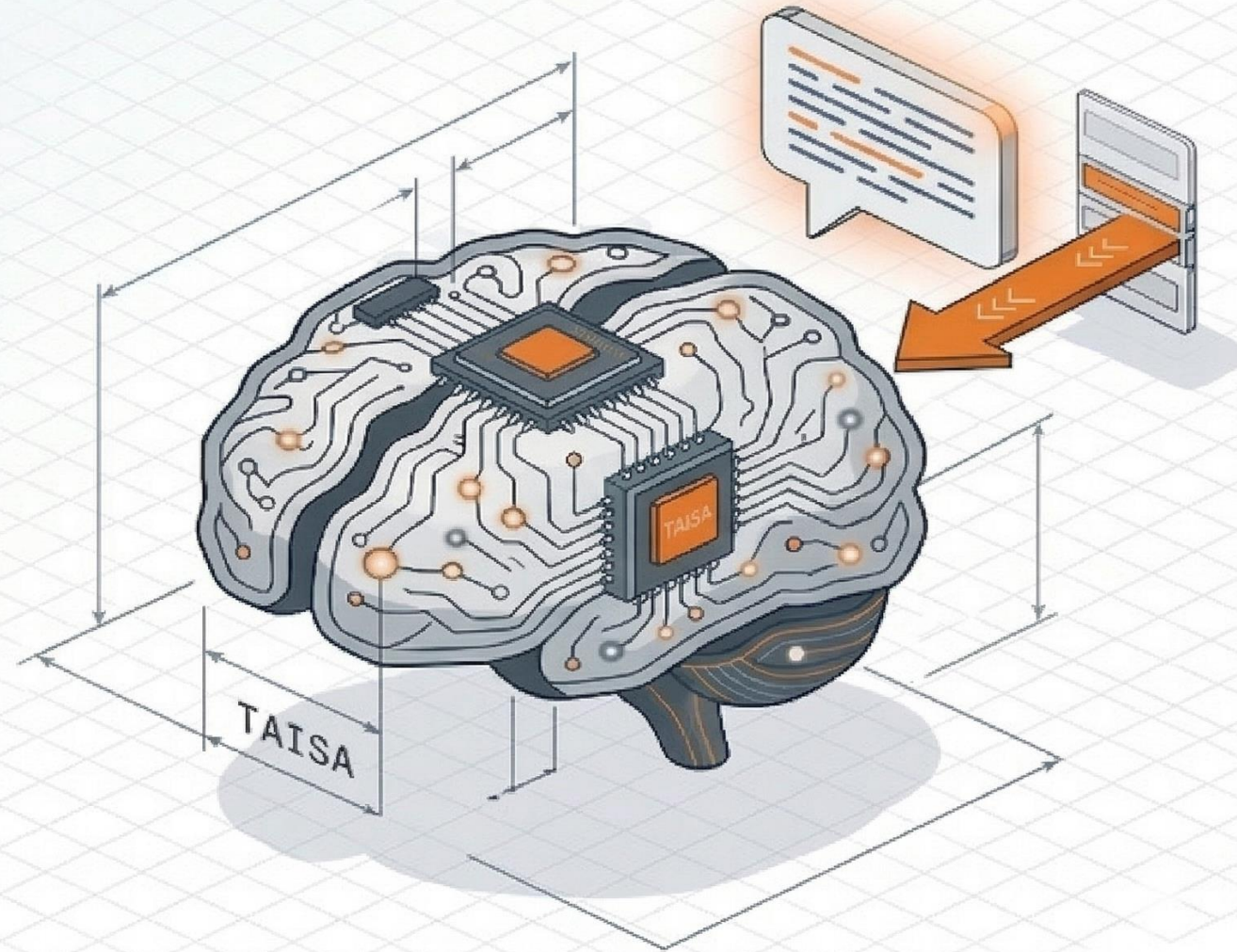
**Role / Purpose:** The AI Augmentation. Interprets structured change and impact events and explains risk in natural language.

**Core Capabilities:**

- Consumes *parser-derived lineage* from the platform.
- Structured event interpretation and risk explanation.
- Deterministic context input from versioned system events.

**Technical Value:** AI operates as “Read-Only”—it interprets facts but cannot corrupt state.

**Strategic Value:** Deploys internal AI (TAISA) as an always-on expert analyst.





# Estimation



# Infrastructure & Deployment Requirements

## Specs & Runtime

### Minimum On-Prem / VM Requirements:

- Compute: 4–8 vCPU
- Memory: 16–32 GB RAM
- Storage: 200–500 GB (Low footprint)
- Runtime: Docker / Fully Containerized

### Cloud Instance Estimate:

- ~\$200–\$600/month (Region dependent)
- ~\$2,400–\$7,200/year (Estimated)
- Scales horizontally via K8s/Docker Swarm

## Cost Contrast



Zero Licensing Fees. No SaaS Dependency.  
100% Owned Infrastructure.



# Engineering Transparency Required Team Profile & Development Model

Internal Engineering Initiative | Not a Consulting Engagement



## Lead Engineer (1 FTE)

- **Focus:** Architecture, Core Logic (Diff/Graph), State Management
- **Responsibility:** System integrity and critical path analysis



## Backend Engineer (1 FTE)

- **Focus:** API Orchestration, Ingestion Adapters, Optimization
- **Responsibility:** Scalability and data flow.

## Required Developer Knowledge

- **Programming:** Python, SQL, (Java/Go optional)
- **Architecture:** Microservices, API Design (REST/gRPC), Event-Driven
- **Data & Logic:** Graph Databases, Diff Algorithms, State Management
- **Infrastructure:** Docker, Kubernetes, CI/CD pipelines.

## UI Development Plan

| Phase       | Activity                | Output             | Tools              |
|-------------|-------------------------|--------------------|--------------------|
| Design      | Wireframing, UX Flows   | Mockups, Prototype | Figma              |
| Development | Frontend Implementation | Responsive Web App | React/Angular, CSS |
| Integration | Connect to APIs         | Functional UI      | REST/GraphQL       |
| Testing     | UAT, Bug Fixing         | Stable UI          | Jest, Cypress      |

## Internal Continuity

Development is owned by Teradata DNA. Source code and architectural knowledge are retained internally, eliminating "black-box" dependency.





# Delivery Timeline & Execution Scenarios

Scenario A: High Velocity (Recommended) | 2 Engineers | 5-6 Months Timeline

## Lead Engineer (1 FTE)



### Focus:

- Architecture, Core Logic (Diff/Graph), State Management.

### Key Tasks:

- Define system architecture & patterns.
- Implement core diffing & graph logic.
- Manage state consistency.
- Perform critical path analysis.

## Backend Engineer (1 FTE)



### Focus:

- API Orchestration, Ingestion Adapters, Optimization.

### Key Tasks:

- Develop API orchestration layer.
- Build ingestion adapters for data sources.
- Optimize performance & scalability.
- Manage data flow and pipelines.

## Milestones:



**M2:** Shadow Mode MVP (Risk Reduction)



**M6:** DNA-Ready Release



# Delivery Timeline & Execution Scenarios

## Scenario B: Low Velocity | 1 Engineer | 9–12 Months Timeline

### Full-Stack Engineer (1 FTE)



#### Focus:

- End-to-end ownership (Architecture + Backend + Integration)
- Incremental delivery & prioritization

#### Key Tasks:

- Define system architecture & patterns
- Implement core engines (Snapshot, Diff, Graph)
- Build API orchestration layer
- Develop ingestion adapters
- Integrate TAISA reasoning
- Manage performance and scalability

#### Milestones:

🚩 M3–M4: Core Engines MVP (Snapshot + Diff)

🚩 M9–M12: Lineage & Impact Ready

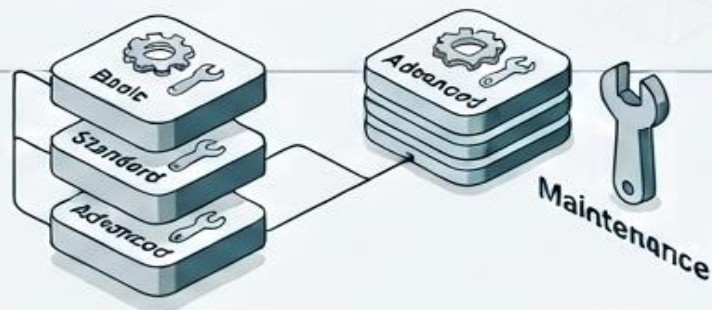
🚩 M6–M8: Lineage & Impact Ready

🚩 M9–M12: DNA-Ready Release



# SCION Maintenance Model & Effort Estimation

Flexible maintenance model aligned with enterprise service tiers



| Capability                  | Basic<br>(Run & Secure) | Standard<br>(Sustain & Adapt) | Advanced<br>(Evolve) |
|-----------------------------|-------------------------|-------------------------------|----------------------|
| Security patches            | ✓                       | ✓                             | ✓                    |
| Dependency updates          | ✓                       | ✓                             | ✓                    |
| Critical bug fixes          | ✓                       | ✓                             | ✓                    |
| Compatibility updates       | ✗                       | ✓                             | ✓                    |
| Minor performance tuning    | ✗                       | ✓                             | ✓                    |
| Environment adaptation      | ✗                       | ✓                             | ✓                    |
| New features / enhancements | ✗                       | ✗                             | ✓                    |
| TAISA evolution             | ✗                       | ✗                             | ✓                    |

## Notes

Recommended: Standard Maintenance for production environments  
Effort estimates can be translated into cost based on internal resource rates  
Estimates assume stable architecture and no major scope changes  
TAISA Integration is non-blocking and independently maintainable

## Maintenance Effort & Tech Profile Estimation (Monthly)



| Tier     | Estimated<br>Effort   | Required<br>Technical Profile                                                                                                                        |
|----------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Basic    | 8-12h<br>hours/month  | <b>Infrastructure Operations Specialist</b><br>Monitors system health, applies routine updates, and manages common incidents.                        |
| Standard | 16-24h<br>hours/month | <b>Senior Platform Engineer</b><br>Optimizes system performance, manages security configurations, and adapts the infrastructure to new requirements. |
| Advanced | 40+h<br>hours/month   | <b>Principal Architect</b><br>Defines the technical roadmap, designs complex new features, and leads the strategic evolution of the platform.        |



# Summary



# SCION – Consolidated View

Integrated overview of architecture, delivery model, and operational footprint

## Architecture



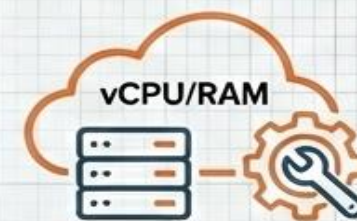
- Modular layered design
- Deterministic processing (Snapshot + Diff + Graph)
- AI augmentation via TAISA (read-only, non-invasive)

## Estimation



- 2 Engineers (1 Lead + 1 Backend)
- 5–6 months delivery timeline
- Shadow Mode MVP by Month 2

## Infrastructure & Maintenance



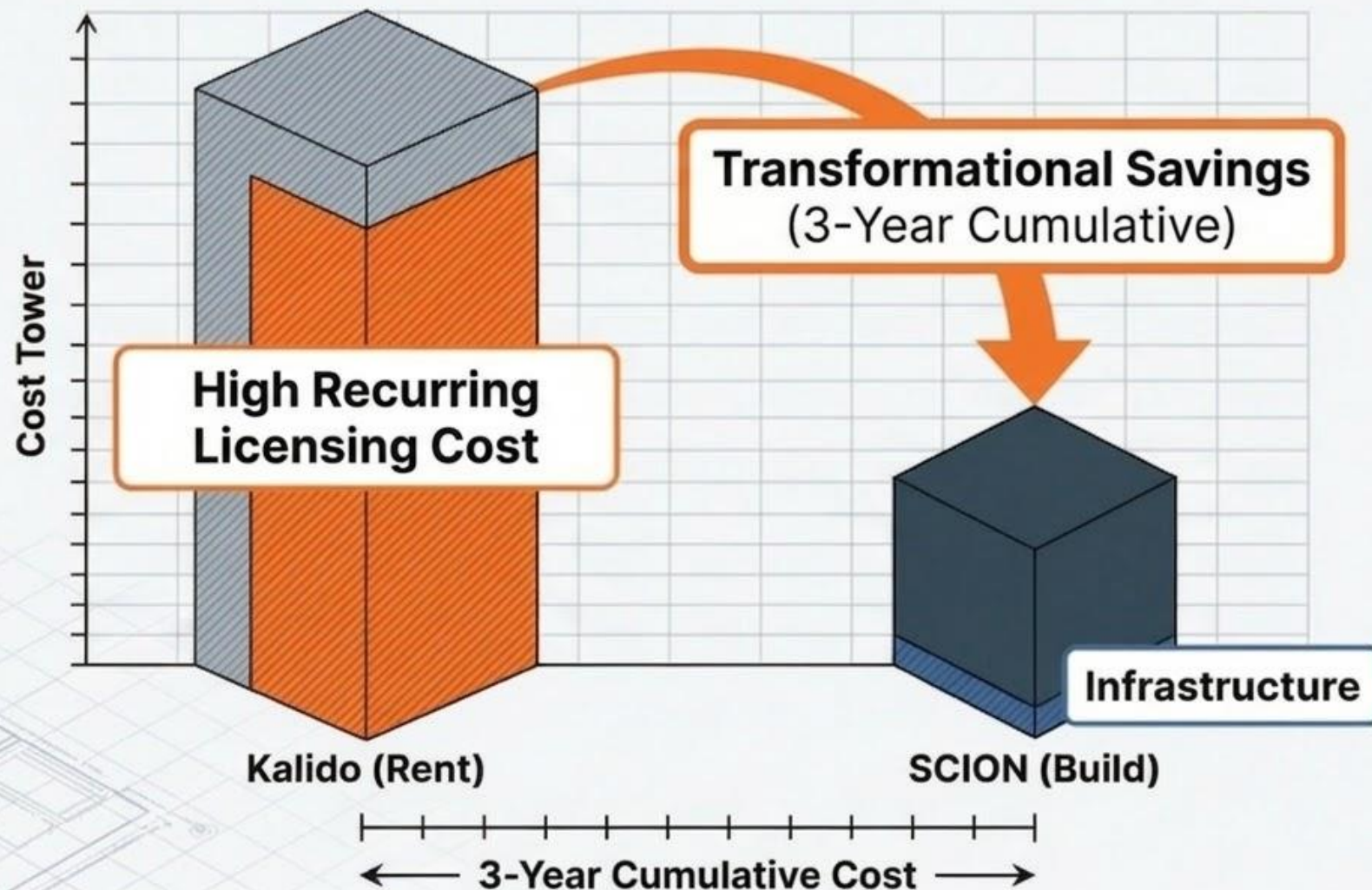
- Low footprint (4–8 vCPU / 16–32 GB RAM)
- No licensing cost (fully owned platform)
- Flexible maintenance model (Basic / Standard / Advanced)

SCION enables a transition from vendor dependency to fully owned structural intelligence.

# Final Recommendations



# Financial & Strategic Impact



- **Immediate:** Eliminates recurring vendor cost.
- **Asset:** Shifts spend from Renting (OpEx) to Building (IP).
- **Risk:** Eliminates “black-box” dependency.

# Phase 2 Strategic Potential (Post-DNA Ready)



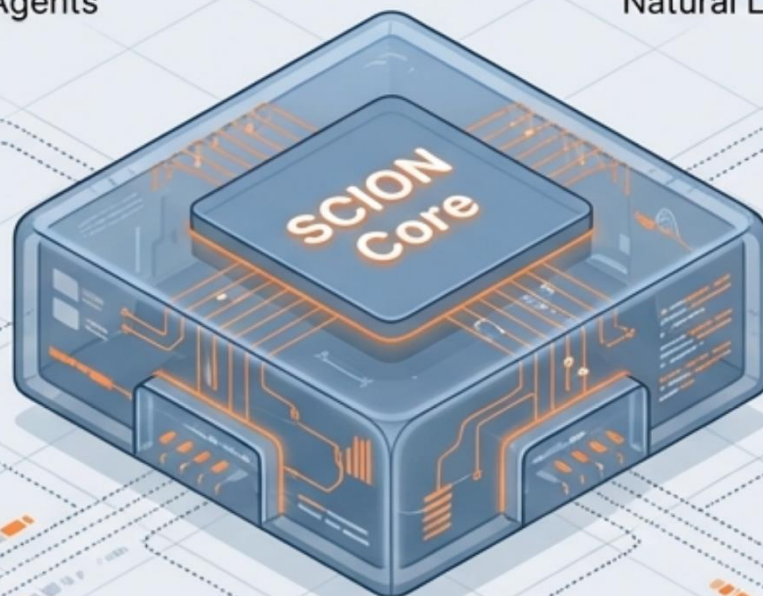
**Intelligent Agent Orchestration**  
Design & Risk Agents

**Conversational Copilot**  
Natural Language Interface



**Predictive Analytics**  
Volatility Forecasting

**Autonomous Advisory**  
Anti-pattern Detection





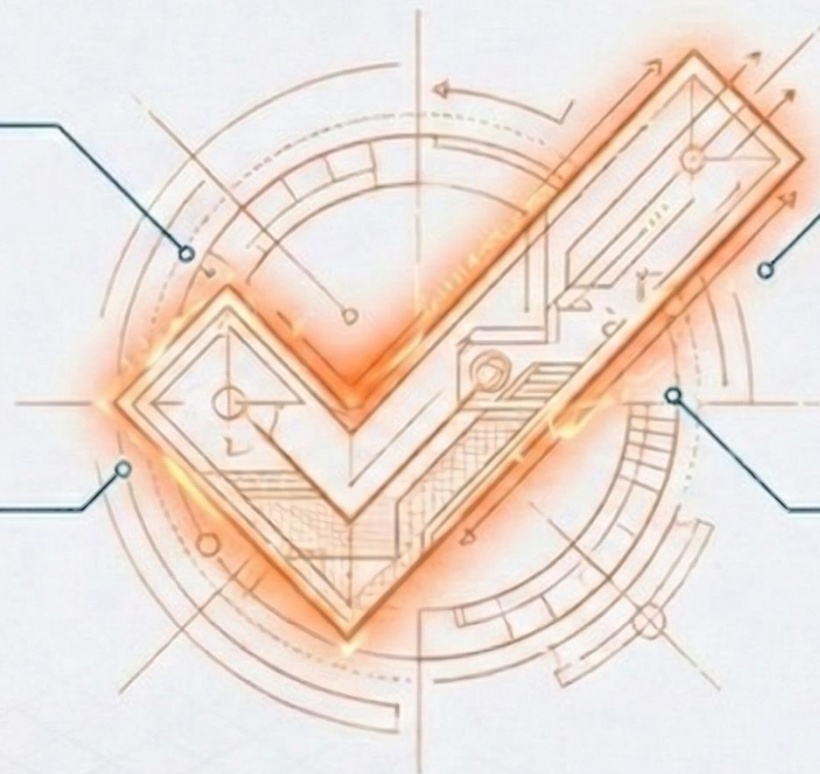
# Executive Recommendation

## Financially Justified:

Clear ROI and elimination of substantial recurring platform expenses.

## Technically Viable:

MVP validated  
Snapshot/Diff logic.



## Architecturally Sound:

Modular, AI-ready design.

## Strategically Essential:

Moves DNA to proprietary independence.

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